

Nathan Zimmerman

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WORK HISTORY

Software Engineer II, Nelnet

10/2023 – Present

- Built core features within a greenfield Payment Plans & Billing domain for a multi-tenant SaaS platform serving educational institutions, spanning C#/.NET microservices, Angular, MongoDB, and shared service integrations.
- Designed payment plan offering and enrollment workflows across admin and payer-facing apps, supporting plan setup, payment rules, schedule scenarios, disclosures, and final enrollment.
- Developed a reusable payment schedule calculator packaged as a shared Angular/npm component, generating installment scenarios from rules around down payments, cadence, fees, and completion dates.
- Integrated payment, banking, disclosure, pricing, and messaging services to support payment method selection, disclosure rendering, enrollment confirmations, and payment plan lifecycle updates.
- Built event-driven returned-payment handling that applied return policy actions including installment rescheduling, failed-payment handling, plan cancellation, and runtime status derivation.

Software Development Engineer, Amazon

05/2022 – 04/2023

- Engineered risk-guidance applications using Java, Scala, Spring, and AWS to screen third-party sellers against denied-party lists and financial crime risk signals, including money-laundering indicators.
- Remediated critical security issues by enabling encryption for SQS/SNS/S3 via KMS, tightening IAM policies to least privilege, and upgrading vulnerable dependencies.
- Executed EC2 host migrations across IAD/PDX Availability Zones for 10+ applications and 100+ environments, validating cutovers with alarms and dashboards to maintain zero downtime.
- Reduced operational support load by resolving pipeline failures, monitoring alarms, and deployment errors across production application workflows.

Software Developer, Radian Group Inc.

09/2019 – 04/2022

- Contributed to the launch and maintenance of MyRadian, a React and Node.js application suite for valuation and asset-management services, supporting feature delivery, reviews, bug fixes, and production support.
- Troubleshoot production issues and improved performance using AWS CloudWatch, X-Ray, DynamoDB, Splunk, and SQL across modern and legacy C#/.NET and Java applications.
- Automated database reconciliation workflows with Python scripts, eliminating recurring manual operations and improving data accuracy.

PROJECTS

- [Sudoku](#) - Built a full-stack Sudoku app with a .NET Web API puzzle generator and Angular gameplay UI. Added validation, rate limiting, and CI-backed tests to keep gameplay reliable. | sudoku.nathanzimmerman.com
- [Blackjack](#) - Built a session-backed Blackjack game with Spring Boot + Spring Security and a React UI. Implemented rules (splits, insurance) plus validation and security hardening. | blackjack.nathanzimmerman.com
- [Nerdle](#) - Built a Wordle-style game with a Node/Express backend that validates guesses server-side. Added rate limiting, theming, and stats tracking. | nerdle.nathanzimmerman.com
- [Brick Breaker Resume](#) - Built a Phaser game that parses a live .docx resume to generate levels from text. Implemented mapping logic, responsive scaling, and score persistence. | resume.nathanzimmerman.com

SKILLS

C#, .NET, Azure, AWS, REST APIs, Microservices, MongoDB, JavaScript, Node.js, Angular, React, Java, CI/CD, Git

EDUCATION

AS, Computer Science, Delaware County Community College, 2018

BA, Psychology, (History minor), Rutgers University School of Arts and Sciences, 2016